

# Measuring the External Field

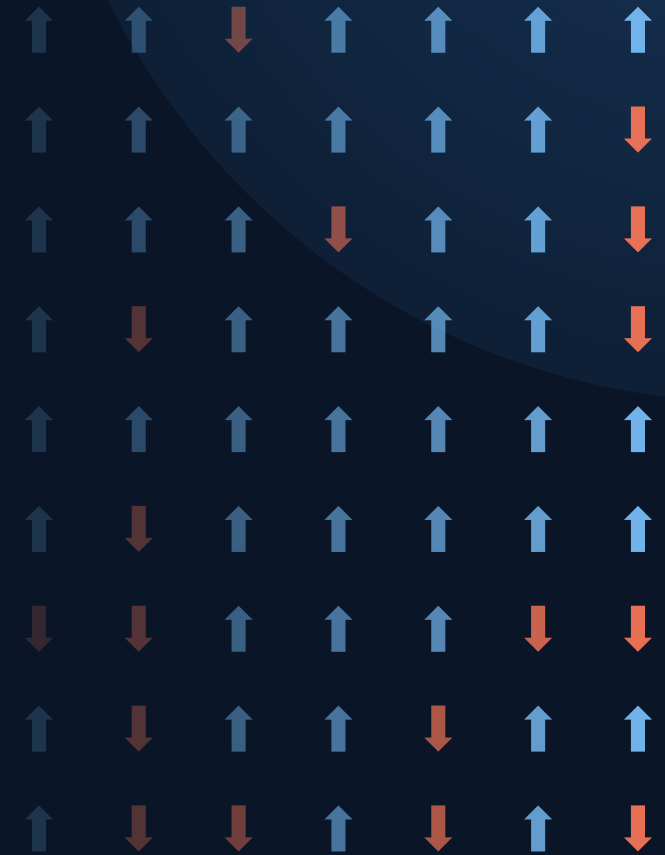
Can we measure how hard a news article pushes on what people think?

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**Brian Mwai** 1050555

Supervisor: Dr. Songa Mutambi

21 September 2026



each arrow is one person's opinion



Physics treats media as a push,  $h$ , on opinion. Nobody measures  $h$

$$m = \tanh(\beta J m + h)$$

 $m$ 

magnetisation  
= average opinion, -1 to +1

 $J$ 

coupling  
= pull toward neighbours

 $\beta$ 

inverse temperature  
= how firmly people follow

 $h$ 

external field  
= media's push on everyone



**Question:** can  $h$  be measured from the content itself?



# Testing the tools changed five assumptions

## WE PLANNED

❌ Measure the amygdala

❌ Score as a ratio

❌ Outrage stands out most

❌ Measure  $\alpha$ , media's strength

❌ A manipulation detector

## WE FOUND

✅ Model sees brain surface only, so we used surface areas

✅ Ratio broke on 69 of 400, so we used a difference

✅ Fear stands out; outrage lifts both areas equally

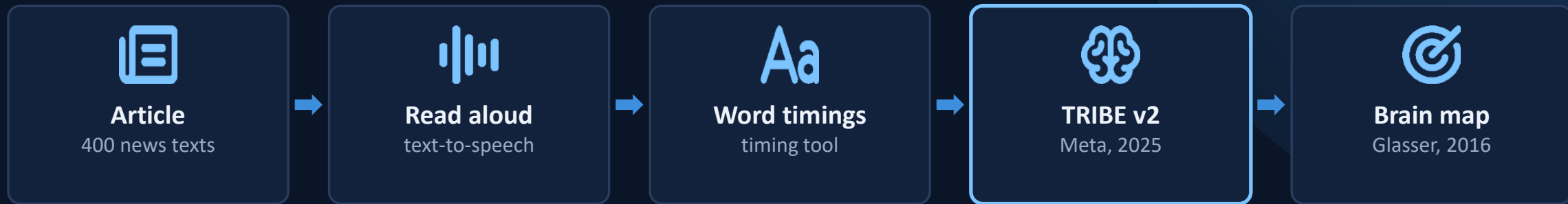
✅ Data can't pin  $\alpha$ , so we derived its minimum

✅ Word counting wins, so it's a measurement

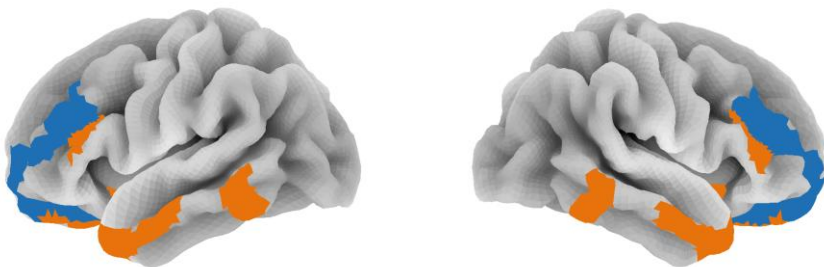
All five are in the approved amendment.



# How one article becomes one number



ROI definition, not a measurement: affective-salience (1030 vertices, orange) and deliberative-control (851, blue)  
left right



orange 1,030 emotion points blue 851 reasoning points

## THE SCORE

$$X = \bar{A}_{\text{emotion}} - \bar{A}_{\text{reasoning}}$$



Nobody scanned: every value is predicted



## 400 articles, four groups, matched in length

**100**

**Fear-driven**

fake news (ISOT)

**100**

**Outrage**

partisan news (SemEval)

**100**

**Clickbait**

headlines (Webis)

**100**

**Neutral**

medical + real news

**≈ 164**

words per article,  
all groups within 5 words

**0.027**

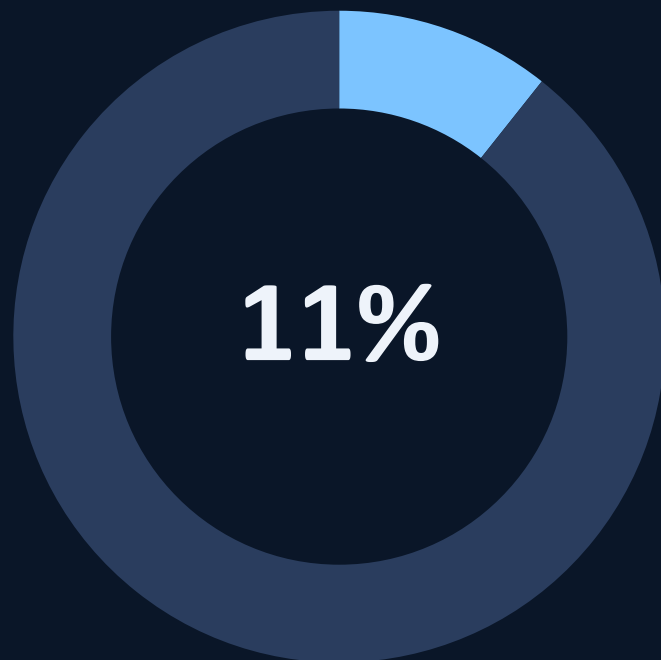
smallest effect we  
could detect, set in advance

**× 2**

every article  
scanned twice



## Result 1: the four groups really differ



11% of the differences between articles comes from their group

$$\eta^2 = \frac{\text{between groups}}{\text{total}} = 0.107$$

$\eta^2$  ON EACH RUN, AGAINST THE SMALLEST WE COULD DETECT



**1 in a billion**

chance it is luck ( $p = 1 \times 10^{-9}$ )

**Repeats**

second run agrees (0.089)

**Meaning:** the groups differ for real, but most variation is from one article to the next.

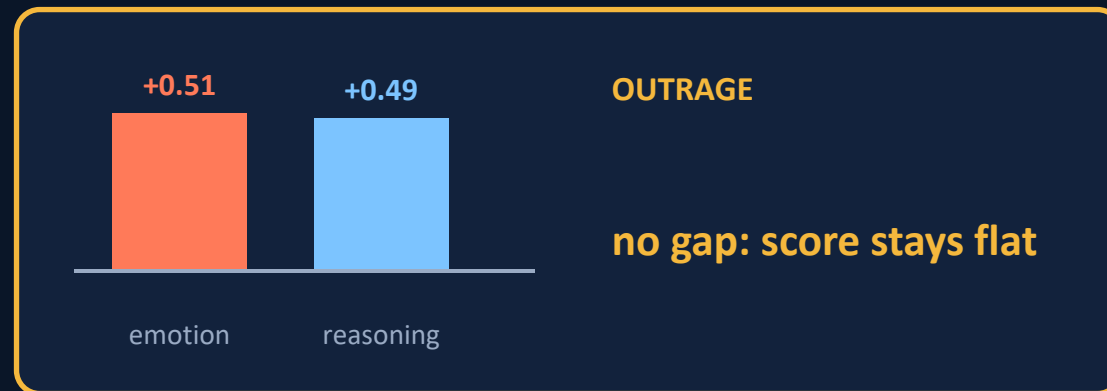
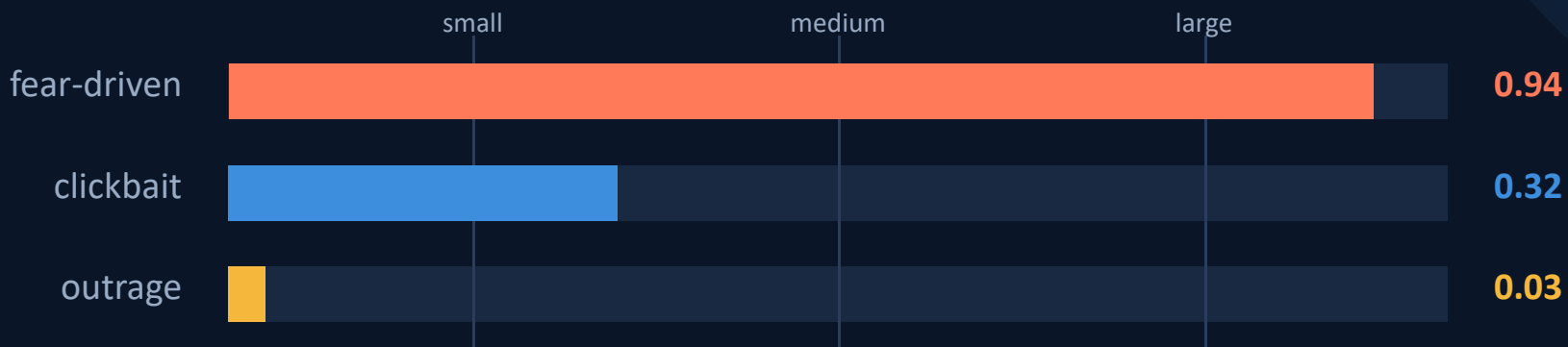


## THE RESULTS

# Result 2: fear drives it; outrage lifts both sides equally

$$d = \frac{\bar{X}_{\text{group}} - \bar{X}_{\text{neutral}}}{s}$$

distance from neutral, in spreads (s)





## Result 3: scores repeat for groups, not single articles

# 0.87

agreement between two runs (ICC)



0 none

1 identical

# 51 of 400

flipped sign between runs



• saw 51 | noise predicts 55 (shaded 44 to 67)

**Meaning:** trust group averages. Never judge one article: its score sits near zero and noise can tip it.

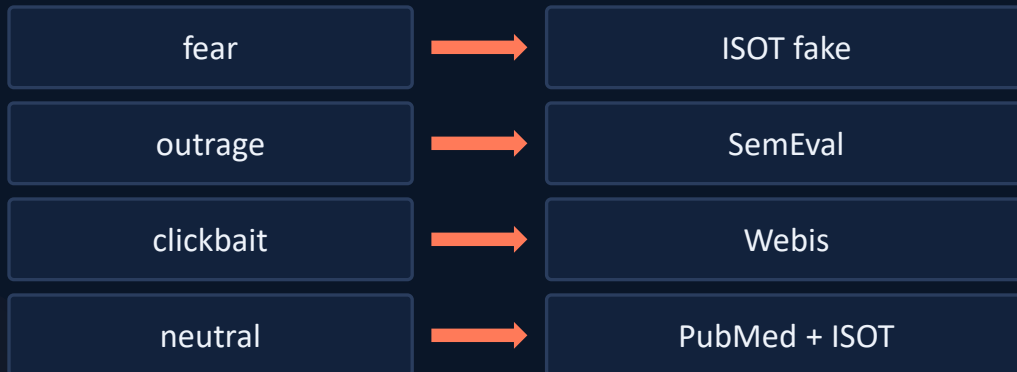


# Honest check: word counting beats our score

How well each method sorts manipulative from neutral articles (AUC)



WHY: EACH GROUP CAME FROM ONE SOURCE



66%

words guess the source  
(chance: 25%)

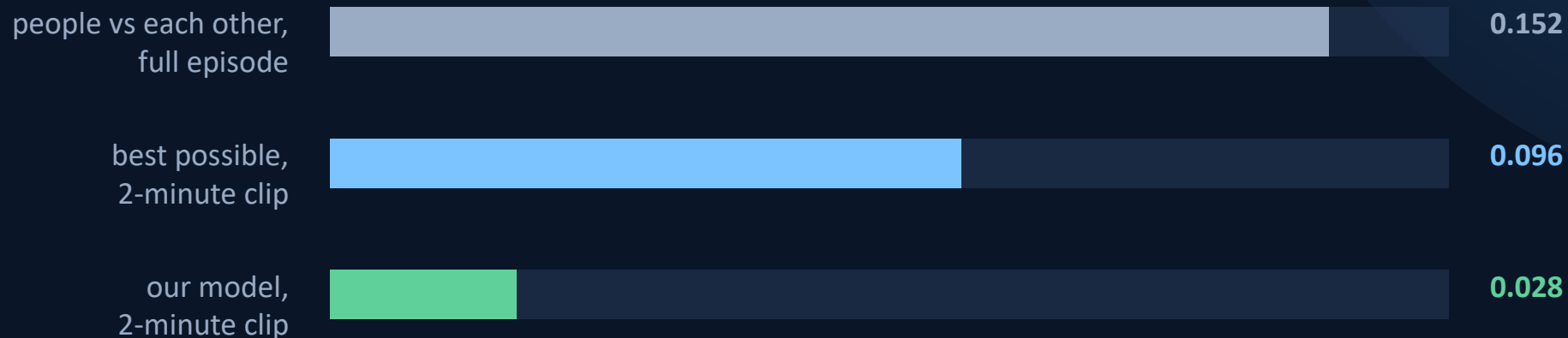
So our score is a  
measurement for the  
physics, not a detector.



## THE RESULTS

# Does the brain model match real brains? Weakly, yes

Tested on public brain scans of people watching Friends (Algonauts 2025). Higher = closer match (r).



≈ 30%

of the best possible on the clip

$p = 0.048$

just under the 5% cut-off

3 bugs

found and fixed along the way

**Meaning:** weak but positive, not the opposite of real brains. A longer run will settle it (Paper 3).



# The minimum push media would need

$$h = \alpha X \Rightarrow \alpha \geq \frac{h_c(\beta J)}{\Delta X}$$

$h$

**external field** = media's push on everyone

$X$

**observable** = our score for an article

$\alpha$

**coupling constant** = push per unit of score

$h_c$

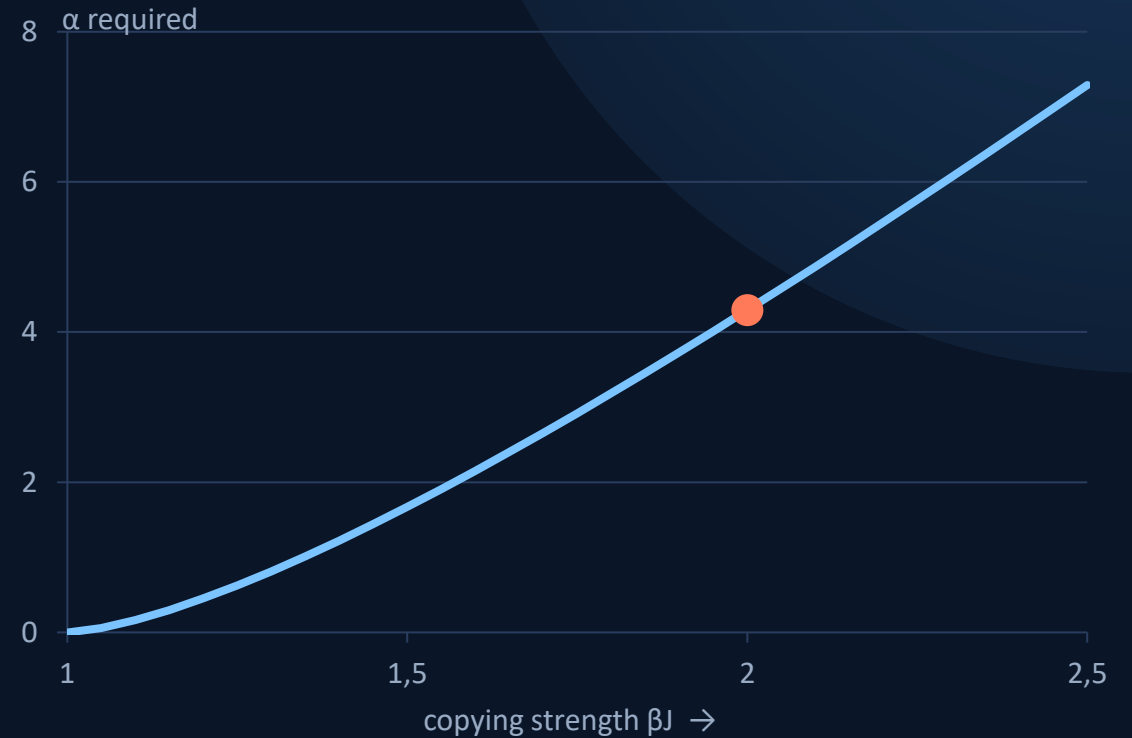
**critical field** = smallest push that flips the majority

$\beta J$

**reduced coupling** = how strongly people copy each other

$\Delta X$

**spread** = range of our scores = 0.124



**$\alpha \geq 4.29$  at  $\beta J = 2$**

a bar any claim must clear



## Can it analyse any content, even what children watch?



### CAN DO

Score video, audio and text

Compare groups of content

Rate content, never the viewer



### NOT YET

Model learned from adult brains

Only news text was tested

One show alone is unreliable



### WOULD TAKE

Child brain data, with ethics approval

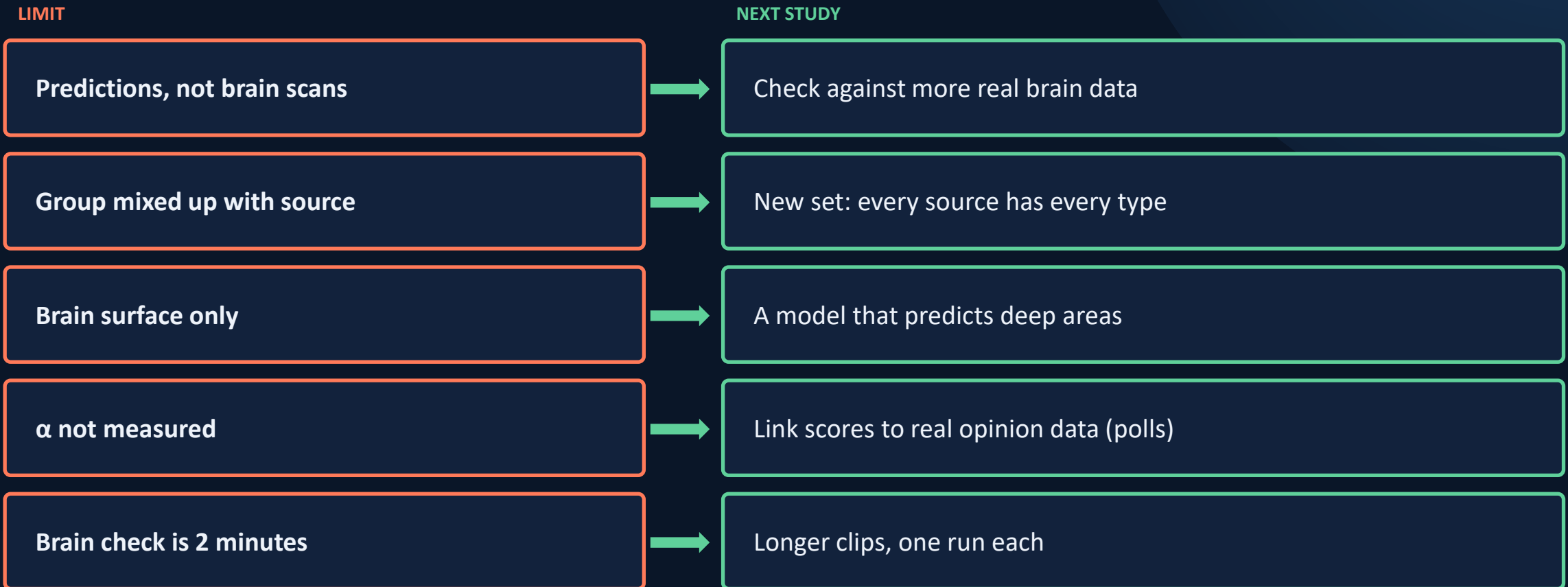
A children's video test set

Safeguards against targeting children

**Short answer:** technically possible, but not yet valid for children.



## Each limit points to the next study





## Taking it further: where this research can go



### PUBLISH

Paper 1 → Physica A

Paper 2 → Physica A

Paper 3 → Imaging Neuroscience

Preprints first (arXiv)



### EXTEND THE SCIENCE

Kenya: 2024 Finance Bill coverage

Video and audio content

Measure  $\alpha$  with opinion data

Children's media, ethically



### OPPORTUNITIES

Postgraduate research topic

Neuroscience + media  
collaborations

Open tool, already online

Group-level media audits

Live instrument: [monarch-4iy.pages.dev](https://monarch-4iy.pages.dev) · Code and data: [github.com/brn-mwai/monarch](https://github.com/brn-mwai/monarch)



## Three papers, and what I need from you today

PAPER 1 · Physica A

### The minimum push

12 pp · ready

PAPER 2 · Physica A

### The measuring tool

16 pp · ready

PAPER 3 · Imaging Neurosci.

### Matching real brains

8 pp · one longer run

## TODAY I AM ASKING FOR



1 Any corrections



2 Signature, page iii



3 Library clearance by Thu  
24



4 Co-authorship view